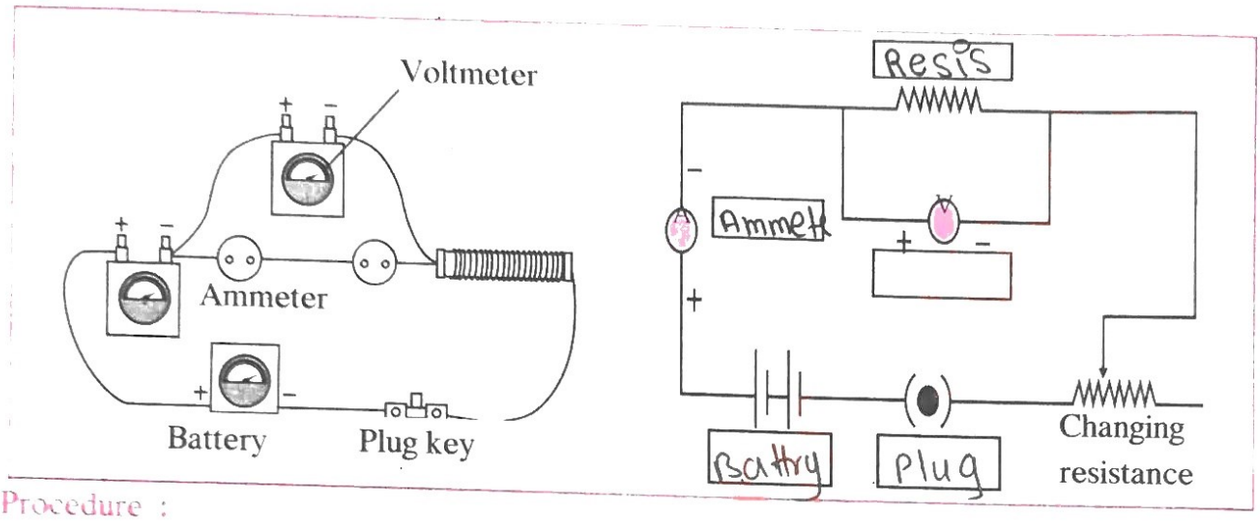
Experiment No. 2

Title : To Study the dependence of the electric current (I) and the Potential difference (V) accross a resistor and determine its resistence and plot a graph of V against I.

Apparatus : Two unknown resistances, Rheostat, Battery, Plug keys, D.C.Voltmeter , D.C. Ammeter, electrical wires, Sandpaper etc.

Figure :



Procedure :

1. Note the zero error and the least count of the voltmeter and ammeter having proper range.
2. Connect the electric circuit as shown in the diagram. Take proper precautions.
3. Close the plug key and allow the electric current pass through the circuit.
4. Adjust the variable resistance to increase the resistance till the expected reading is obtained on the ammeter.
5. Note the voltmeter reading and open the circuit by opening the plug key.
6. Repeat the procedure in the steps 3 and 4 and take minimum 3 to 5 readings.
7. Plot the readings of electric current (I) and the potential difference (V) on the graph paper and draw the graph of V against I.

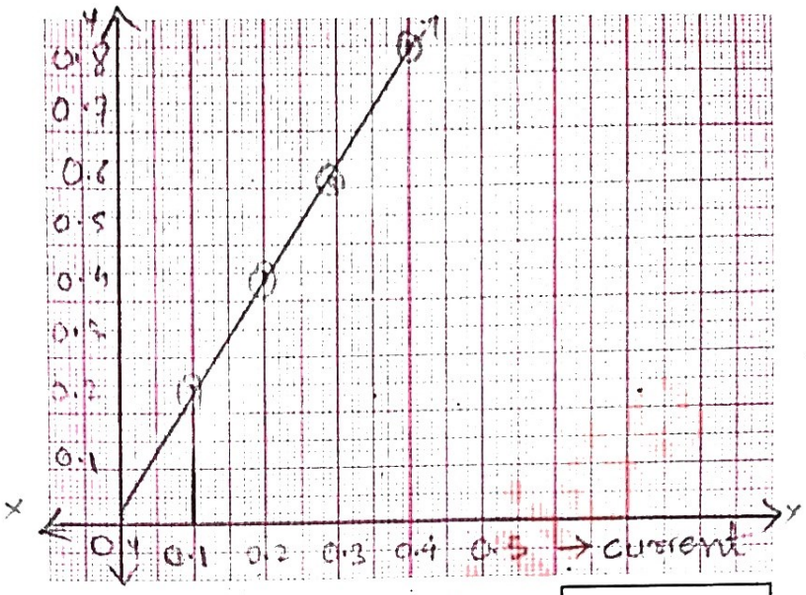
Observation : 1. Least Count of Ammeter =.....A 2. Least count of Voltmeter =.....V
3. Zero error of Ammeter = ...:.A. 4. Zero error of Voltmeter =.....V

Observation Table :

Resistance	Reading	Ammeter Reading (I) Ampere (A)		Votlmeter Reading (V) Volt (v)		$R = \frac{V}{I}$ ohm
		Actual Reading	Corrected Reading	Actual Reading	Corrected Reading	
	1	0.1	0.1	0.20	0.20	2.0
	2	0.2	0.2	0.45	0.45	2.2
	3	0.3	0.3	0.65	0.60	2.2
	4	0.4	0.4	0.80	0.80	2.0
	5	0.5	0.5	1.10	1.10	2.2

Graph :

...Graph of the.....
.....potential.....
...different.....
...and current.....
.....



Calculation :

Mean Value of Resistance
= ..2.12.... Ohms

Inferance :

1. The electric current flowing through the circuit varies directly with the .....
2. Value of unknown resistance = ..2.12....Ohm
3. Graph between (V) and (I) ..is straight line.....

VDFQHT

Multiple Choice Questions

1. When the resistance of the conductor increases then the current will .....
   A. increase ✓B. decrease C. remain same D. becomes double
2. What will happen to current passing through a resistance, if potential difference across it is doubled and the resistance is halved .....
   A. remains unchanged B. becomes double C. becomes half ✓D. becomes four times
3. Straight line is obtained on drawing the graph of current against potential difference. This proves .....law.
   A. Faraday's Law ✓B. Ohm's Law C. Fleming's left hand rule D. Maxwell's Law
4. There are 20 divisions in the range 0 to 0.5 V in a Voltmeter, Therefore least count of the Voltmeter is .....
   A.0.020 ✓B. 0.025 C. 0.050 D. 0.250
5. If the resistance in the circuit is increased four times by keeping the potential difference same, the current in the circuit is .....
   A. remains same B. four times ✓C. one fourth D. half

Questions

1. Which law is proved in the experiment of electric current, potential difference and electric resistance? explain it.

In these experiment Ohm's law proved it is said that the flow current is directly proportional the potential different between on the two end.....

2. What will be the effect of change in the length and thickness of the electrical wire in the experiment of electric current, potential difference and electric resistance?

If the length and thickness of electrical wire with change then potential difference remains same and electric resitance get change.....